ASR Model Selection Report

Egyptian Arabic speech recognition — bake-off results

Prepared 23 July 2026 · Egyptian Arabic voice robot stack

Best robot realtime	Best accuracy	Best speed	Models OK
Whisper-Large-v3-Turbo-CT2	**Whisper-Large-v3-CT2**	**MMS-1B-all**	**12**

Executive verdict

Production default: Whisper-Large-v3-Turbo-CT2 (best accuracy/speed/VRAM blend). Accuracy alternative: Whisper-Large-v3-CT2.

Overview

12 ASR models were evaluated successfully on the Arabic test clip.
Robot realtime score weights: accuracy 45% + speed/RTF 30% + low VRAM 15% + fast load 10%.
RTF (Real-Time Factor) = transcribe_time / audio_duration. LOWER better. RTF < 1 = realtime.
WER (Word Error Rate): LOWER better. Accuracy %: HIGHER better (approx. 1 - WER).
Top composite pick: Whisper-Large-v3-Turbo-CT2.
Top accuracy pick: Whisper-Large-v3-CT2.

How to read this report

Page 2: methodology, test conditions, and limitations.
Page 3: metrics glossary — RTF, WER, CER, VRAM.
Page 4: recommended picks with rationale.
Page 5: full leaderboard evidence table.
Pages 6-10: accuracy, speed, WER/CER, VRAM, and score charts.
Page 11: selection guidance and sign-off checklist.

ASR — Evaluation methodology & scope

What was evaluated

12 models completed transcription successfully (100% success in this bake-off).
Evaluation used the shared Egyptian Arabic test clip from the ASR bake-off suite.
Whisper CT2 models use faster-whisper (CTranslate2) for lower latency and VRAM.
Hardware reference: NVIDIA T4 ≈ 14.9 GB VRAM co-residency budget for ASR+LLM+TTS.

How scores were produced

Robot score = 45% accuracy + 30% speed/RTF + 15% VRAM efficiency + 10% load (HIGHER better).
Accuracy % ≈ (1 − WER) × 100. CER is also reported (important for Arabic orthography).
RTF = transcribe_time / audio_duration; realtime-capable means RTF < 1.
All scores are relative within this candidate set.

Limitations (read before sign-off)

Single-clip bake-off — re-validate on your mic, noise floor, and dialect mix before freeze.
Arabic fine-tunes were competitive on speed but did not beat base Whisper Large on accuracy here.
Qwen ASR / QwenCleo underperformed badly on WER for this Egyptian Arabic clip.
WER/CER depend on reference normalization; treat absolute % as comparative, not absolute truth.

ASR - Metrics glossary (what to prefer)

Use this page while reading the charts. Direction labels show what you want for a production robot.

Accuracy % (word accuracy)

HIGHER better
Share of words recognized correctly (about 1 - WER). Higher = fewer recognition mistakes.

WER - Word Error Rate

LOWER better
Fraction of words wrong. 0 = perfect. Lower = better recognition.

CER - Character Error Rate

LOWER better
Same idea as WER at character level. Useful for Arabic. Lower = better.

Transcribe time (s)

LOWER better
Wall time to finish transcription. Lower = faster handoff to the LLM.

Realtime capable

Prefer YES
Yes when RTF < 1. Needed for near-live listening on the robot.

Peak VRAM (MB)

LOWER better
GPU memory used at peak. Lower leaves room to run ASR + LLM + TTS together on one GPU.

Peak RAM (MB)

LOWER better
System (CPU) memory used at peak. Lower is safer on smaller servers / VPS hosts.

Peak CPU %

LOWER better
CPU utilization spike during the run. Lower means less contention with other services.

Cold load time (s)

LOWER better
Seconds to load the model into memory the first time. Matters for startup / cold starts.

RTF (Real-Time Factor)

LOWER better
Processing time / audio duration. RTF < 1 = faster than realtime (good). RTF > 1 = slower than the audio.

x Realtime (xRT)

HIGHER better
How many times faster than realtime (about 1/RTF). Example: 6x = 6s of audio in ~1s of compute.

Robot realtime score (0-100)

HIGHER better
Composite score for robot use (speed + resources + accuracy where relevant). Higher = better fit.

Balanced score (0-100)

HIGHER better
Normalized blend of the main tradeoffs. Higher = better all-rounder.

Success rate %

HIGHER better
Share of runs that completed without error. Prefer 100% for production reliability.

Remember

For ASR: want HIGHER accuracy %, HIGHER robot score, HIGHER xRealtime;
want LOWER WER, LOWER CER, LOWER RTF, LOWER VRAM/RAM, LOWER load/transcribe time.
Prefer realtime-capable (RTF < 1) models for live robot listening.

Recommended picks (automated)

Best for robot realtime (HIGHER score better)

Whisper-Large-v3-Turbo-CT2

Highest composite robot score (accuracy + speed + VRAM + load).

score_robot_realtime=89.62 | avg_accuracy_percent=68.4 | avg_rtf=0.046 | peak_vram_mb=2492.2

Best accuracy % (HIGHER better)

Whisper-Large-v3-CT2

Highest average word accuracy / lowest WER.

avg_accuracy_percent=72.92 | avg_wer=0.2708 | avg_cer=0.1132

Best speed / lowest RTF (LOWER better)

MMS-1B-all

Lowest average RTF (fastest relative to audio length).

avg_rtf=0.044 | avg_x_realtime=22.88 | avg_transcribe_seconds=5.64

Lowest peak VRAM (LOWER better)

Whisper-Small-CT2

Lowest peak VRAM — useful for 6–16 GB GPUs or multi-model co-residency.

peak_vram_mb=1340.2

Best balanced

Whisper-Large-v3-Turbo-CT2

Balanced accuracy/speed/VRAM tradeoff.

score_balanced=90.17

Best realtime + accuracy

Whisper-Large-v3-CT2

RTF < 1 and best available accuracy among realtime-capable models.

avg_rtf=0.133 | avg_accuracy_percent=72.92

Selection notes

Prefer LOW RTF + HIGH accuracy + LOW VRAM so LLM/TTS still fit on the same GPU.
Whisper CT2 (faster-whisper) models dominate the top of the robot leaderboard.
Arabic fine-tunes were competitive on speed but did not beat base Whisper Large on accuracy here.
Voxtral-Mini has strong accuracy but HIGH VRAM — costly for co-residency.

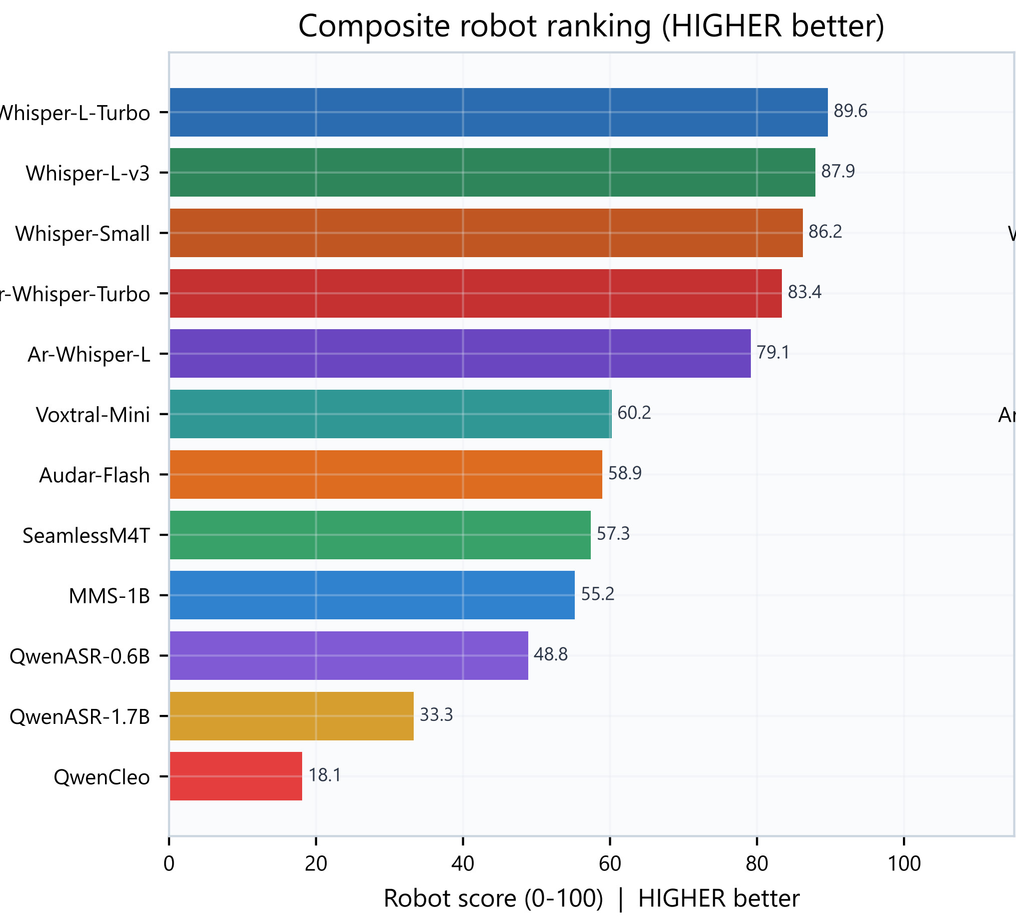
ASR leaderboard (primary evidence table)

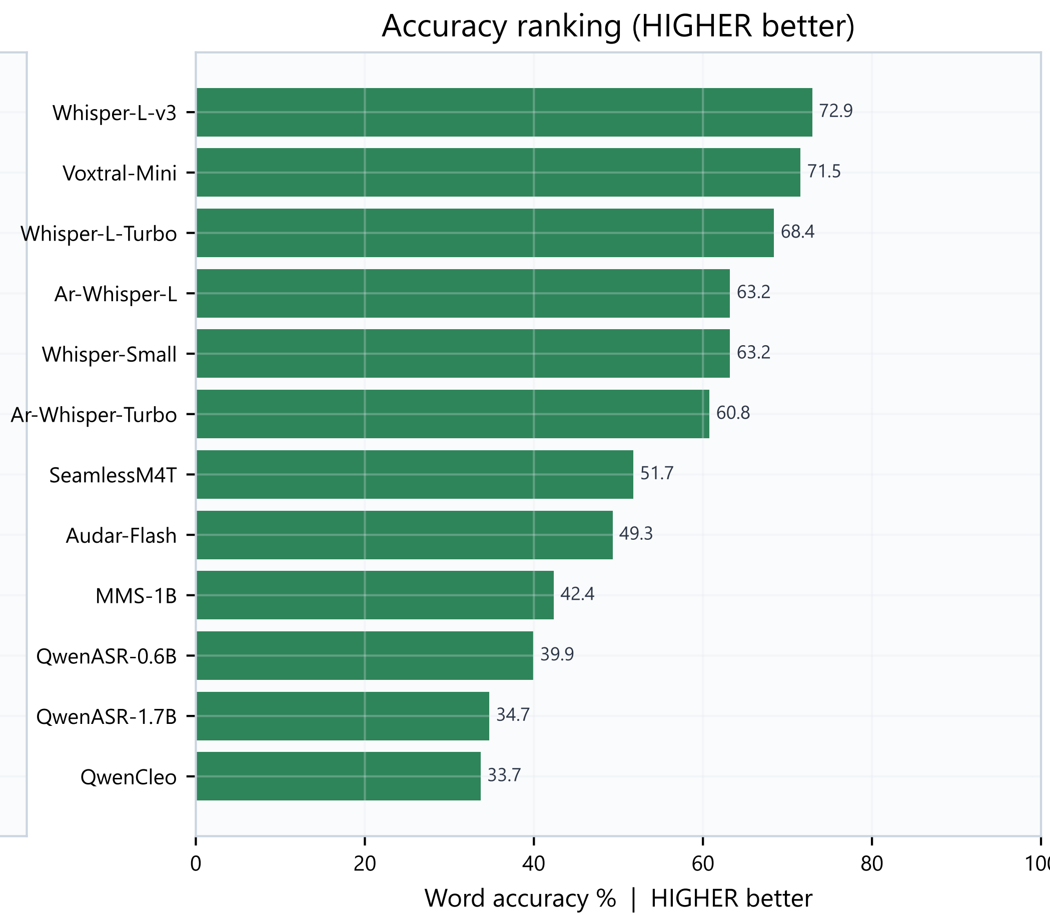
Rank	Model	Robot	Acc%	WER	CER	RTF	VRAM MB
1	Whisper-L-Turbo	89.6	68.4	0.316	0.166	0.046	2492
2	Whisper-L-v3	87.9	72.9	0.271	0.113	0.133	4316
3	Whisper-Small	86.2	63.2	0.368	0.169	0.084	1340
4	Ar-Whisper-Turbo	83.4	60.8	0.392	0.178	0.046	1692
5	Ar-Whisper-L	79.1	63.2	0.368	0.179	0.135	2812
6	Voxtral-Mini	60.2	71.5	0.285	0.103	0.281	11154
7	Audar-Flash	58.9	49.3	0.507	0.413	0.191	3480
8	SeamlessM4T	57.3	51.7	0.483	0.290	0.135	4072
9	MMS-1B	55.2	42.4	0.576	0.270	0.044	5206
10	QwenASR-0.6B	48.8	39.9	0.601	0.401	0.191	3740
11	QwenASR-1.7B	33.3	34.7	0.653	0.533	0.215	6044
12	QwenCleo	18.1	33.7	0.663	0.589	0.582	5870

Reading the table

Sort key is robot realtime score (HIGHER better).
Acc% HIGHER better; WER/CER/RTF/VRAM LOWER better.
Top band is Whisper CT2 family. Bottom band (Qwen ASR/QwenCleo) is not production-ready here.
Use Acc% + RTF + VRAM together — never pick on speed alone.

Robot score & Accuracy

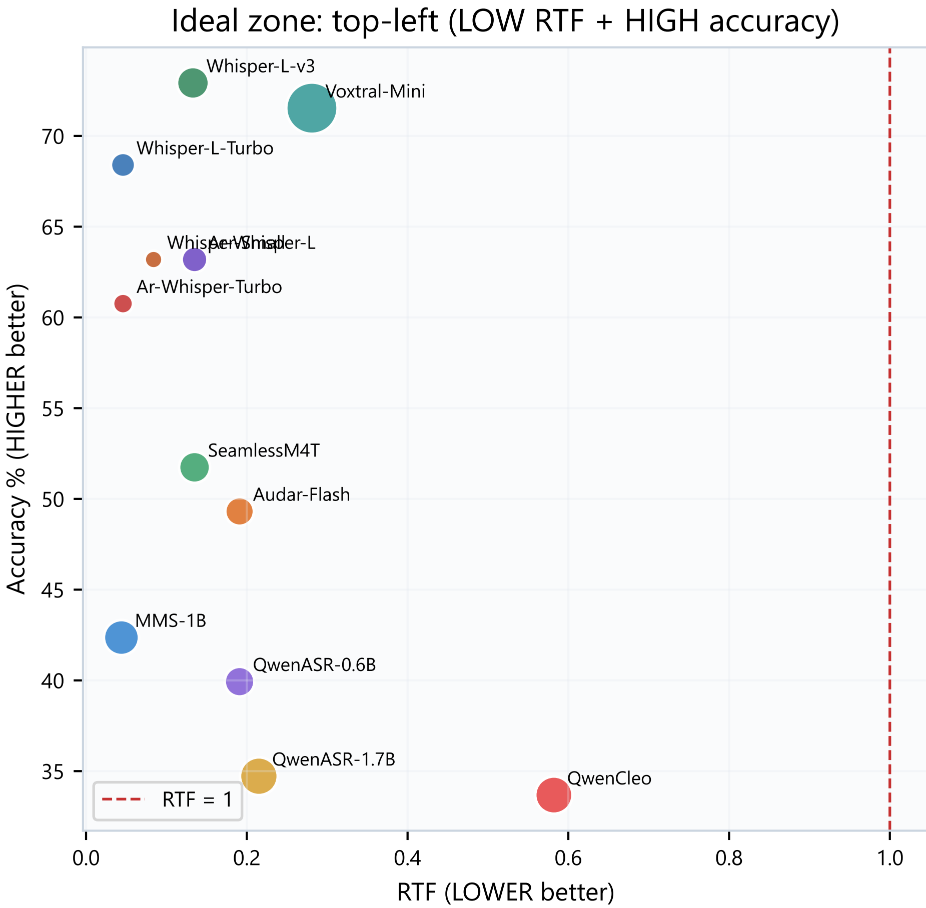


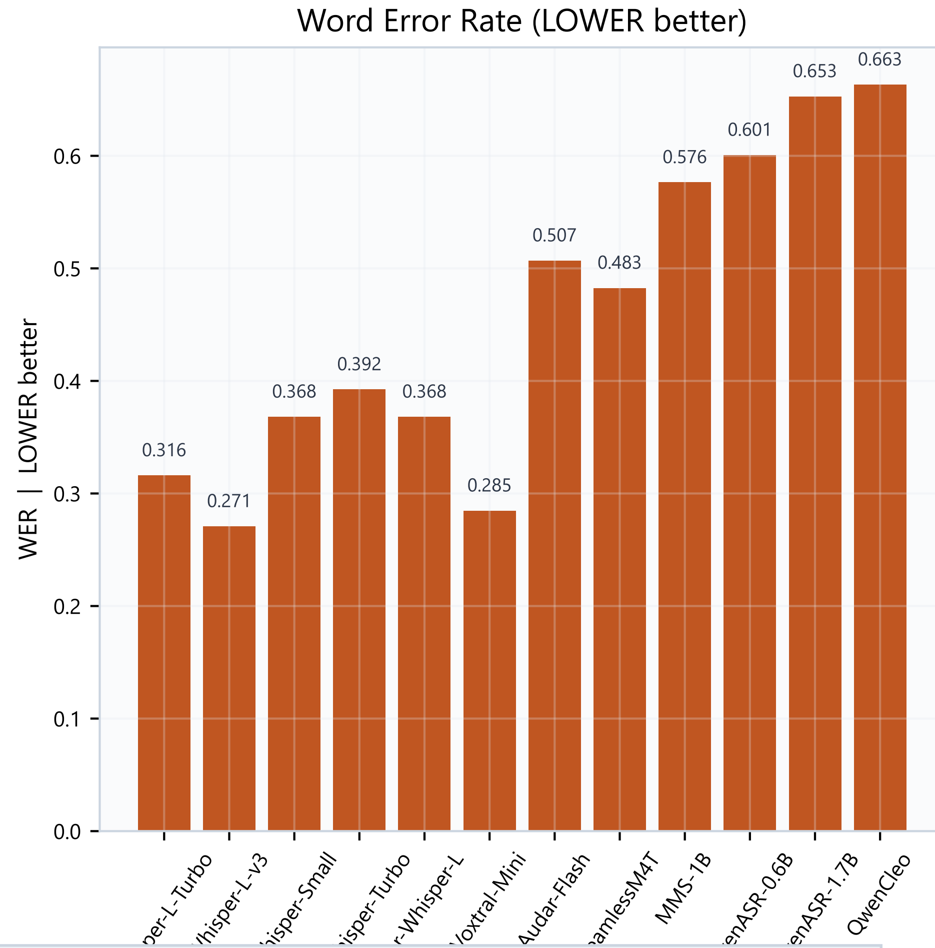


What this means

Left: overall robot fitness (HIGHER better). Right: pure recognition accuracy (HIGHER better).
Whisper-Large-v3-Turbo leads robot score by combining solid accuracy with excellent RTF and moderate VRAM.
Whisper-Large-v3 leads accuracy (72.92%).
Smaller Whisper variants trade some accuracy for LOWER VRAM — useful on tight GPUs.
Bottom of the list (Qwen ASR family / QwenCleo) is too weak on accuracy for production Egyptian Arabic.

Accuracy–speed tradeoff & WER

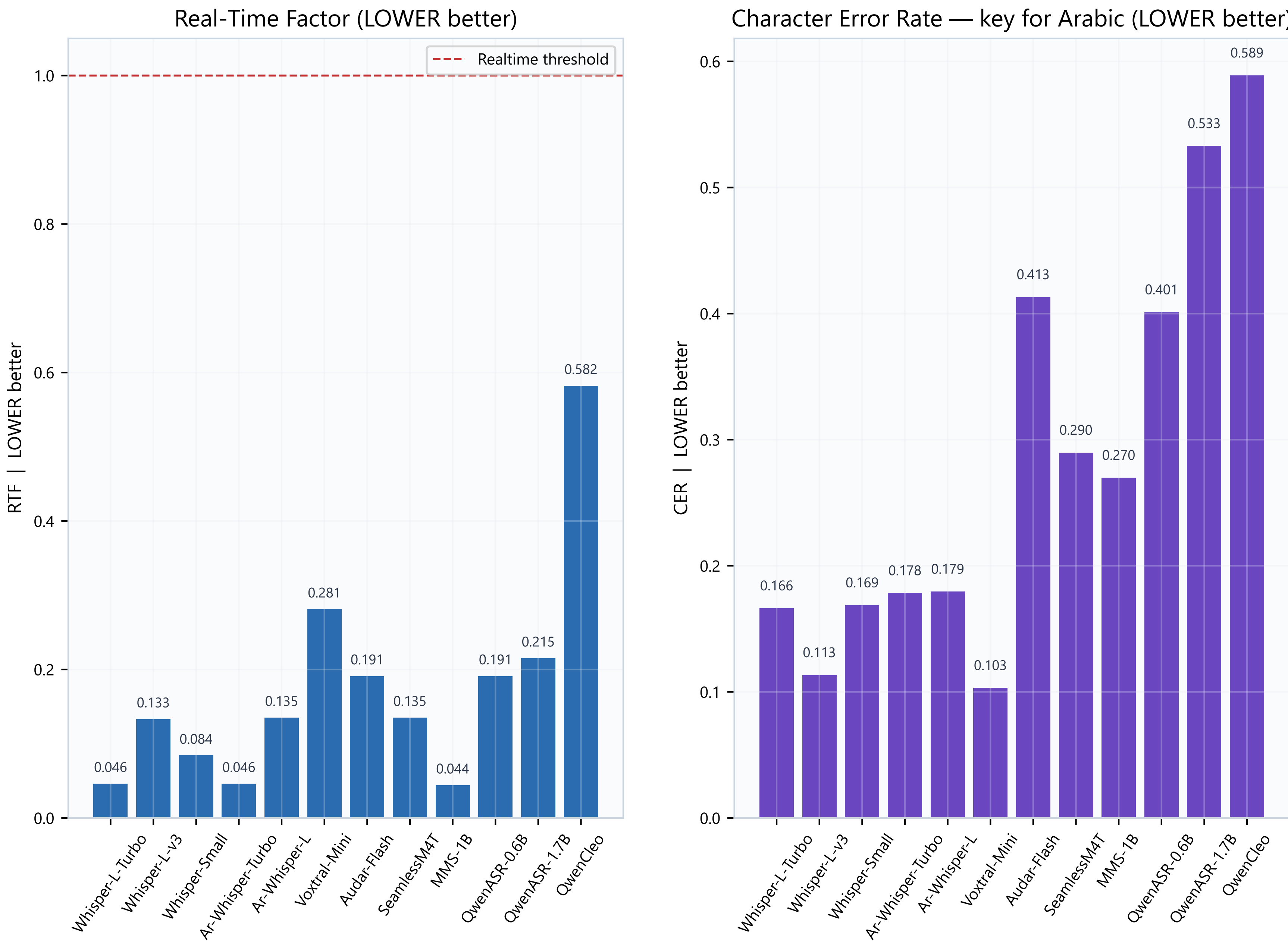




Tradeoff reading

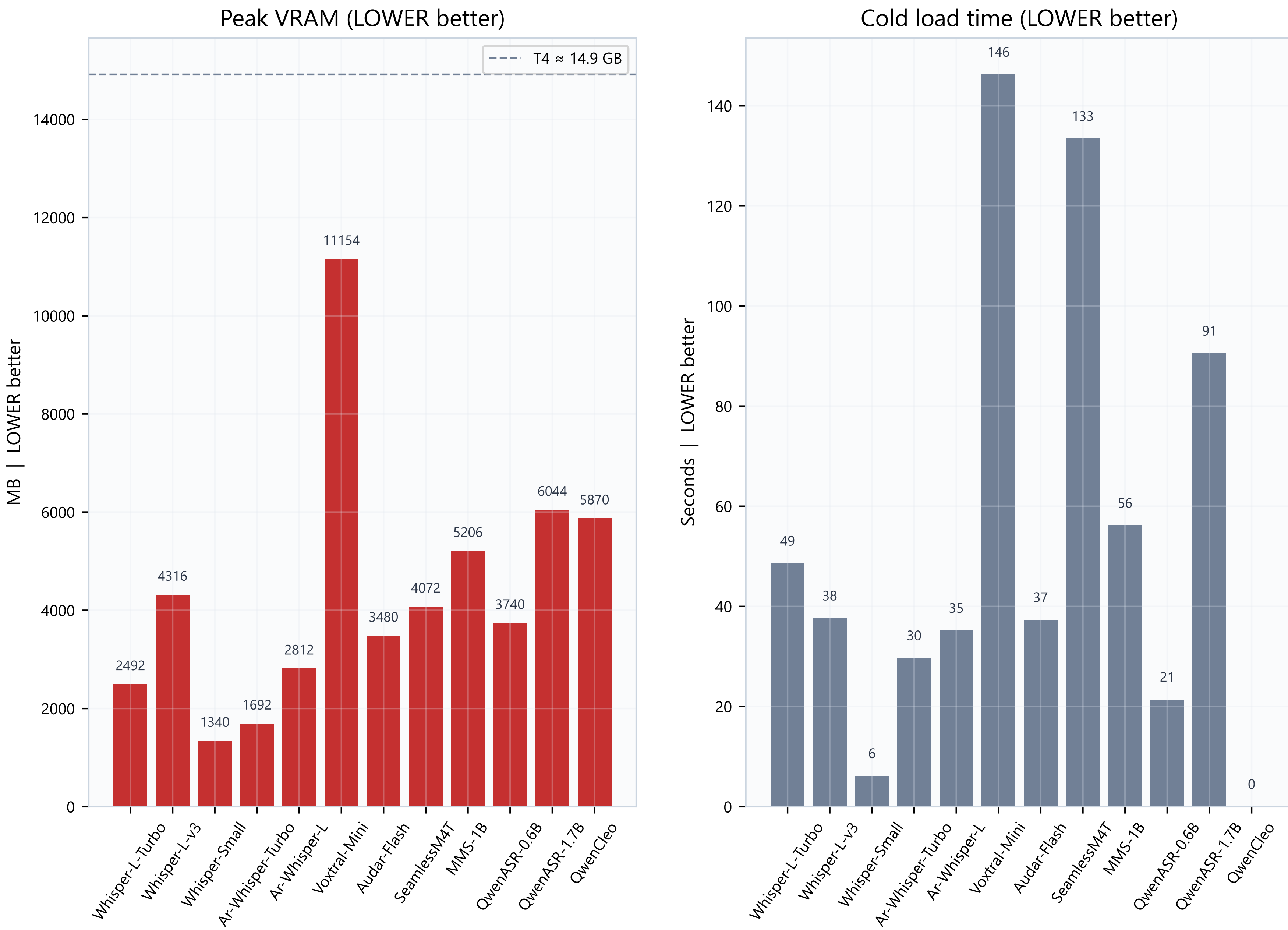
RTF = transcribe_time / audio_length (LOWER better). RTF < 1 = realtime.
WER = Word Error Rate (LOWER better). Accuracy % ~ (1 - WER) x 100 (HIGHER better).
Bubble size = peak VRAM (LOWER better for co-residency with LLM/TTS).
Top-left models are both accurate and fast — preferred for voice robots.
All evaluated models were realtime-capable (RTF < 1) on this clip.
Voxtral is accurate but HIGH VRAM — expensive to host beside an LLM.

Speed metrics & Character Error Rate



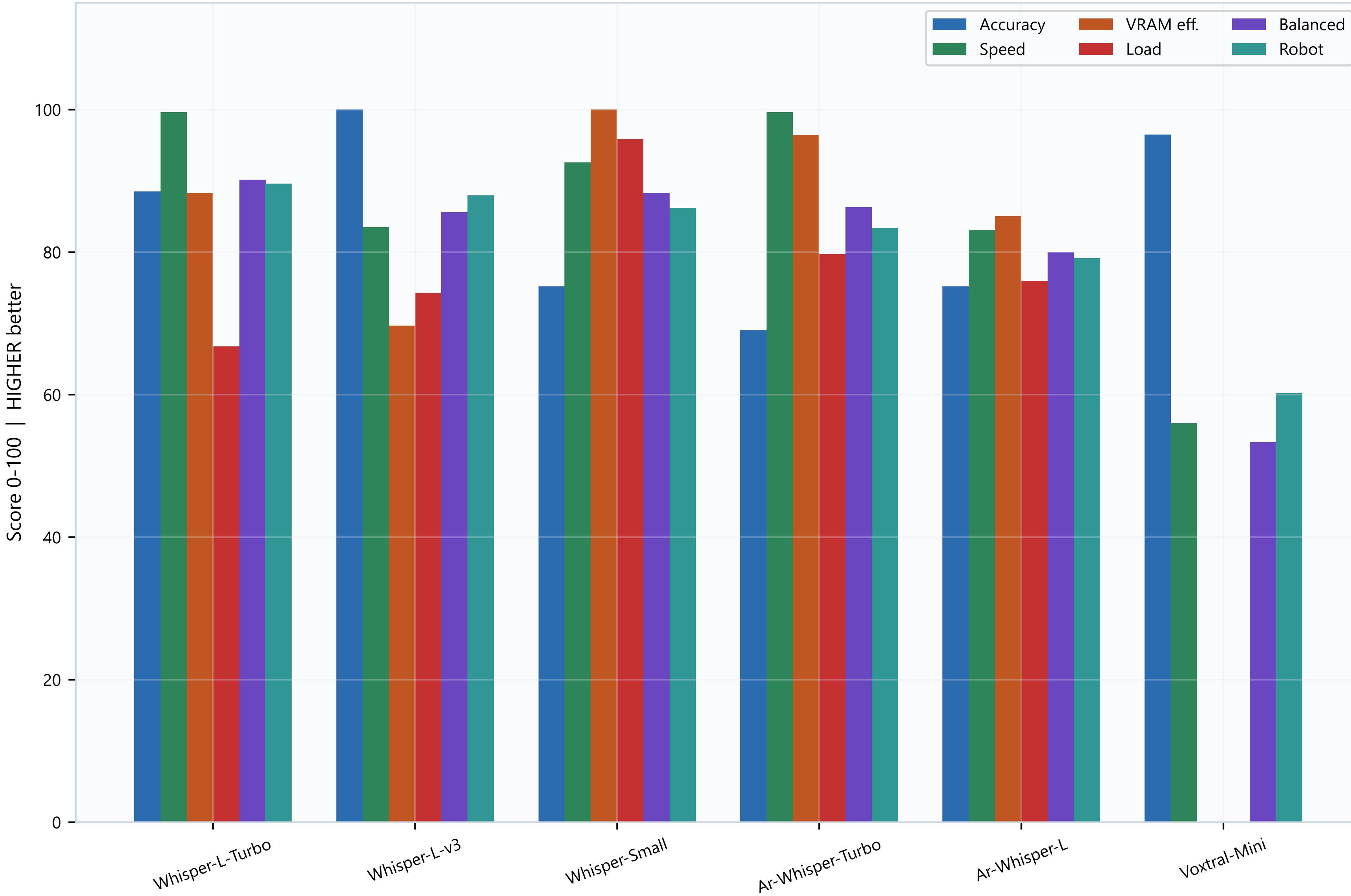
RTF LOWER better (all models < 1 here). CER LOWER better — useful for Arabic where word boundaries are noisy. Whisper-Large-v3 leads CER/WER; Turbo stays close while remaining much faster.

Resources (LOWER better on both charts)



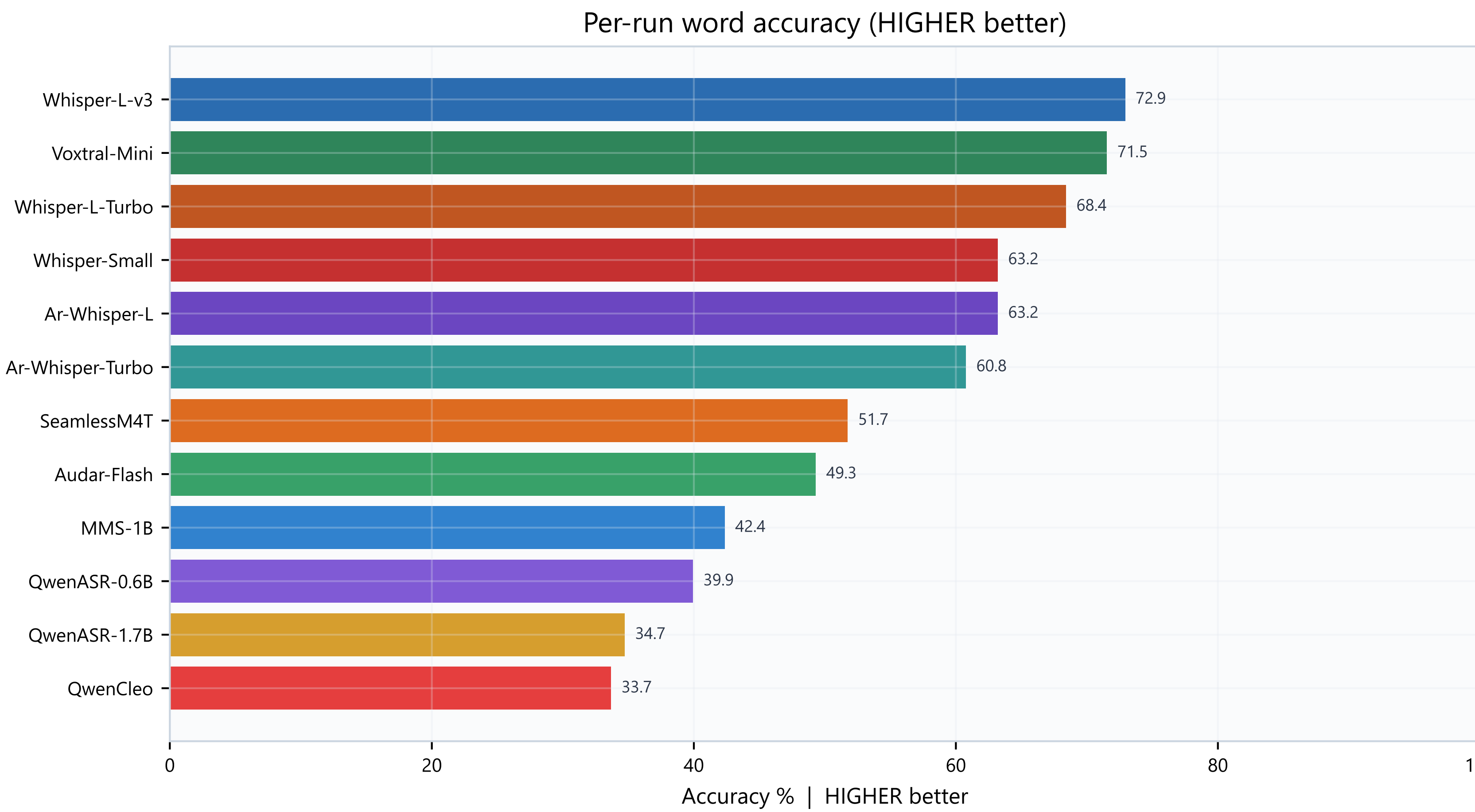
Whisper-Small is the VRAM/load champion (~1.3 GB). Whisper-Large-v3-Turbo stays under ~2.5 GB while ranking #1 on robot score — usually the best default when LLM must share the GPU. Avoid Voxtral (~11 GB) unless ASR runs alone.

Normalized score breakdown (top 6) - all HIGHER better



Use this breakdown when two models have similar robot scores. Example: Turbo wins on speed+VRAM blend; full Large-v3 wins accuracy. Pick based on whether recognition errors or latency hurt your robot more.

Accuracy ranking & selection guidance



How to choose / sign-off checklist

1) Default production pick → Whisper-Large-v3-Turbo-CT2 (best robot composite).
2) If accuracy is the hard constraint → Whisper-Large-v3-CT2.
3) If GPU is tight / multi-model → Whisper-Small-CT2 (lowest VRAM, still decent).
4) Skip Qwen ASR / QwenCleo for this Egyptian Arabic use case based on current WER.
5) Re-validate on your microphone noise and dialect samples before freeze.
6) Confirm combined VRAM with chosen LLM + TTS still fits the target GPU.